

Shaping a Smarter World with AI, Machine Learning and GIS

A Geographical Information System (GIS) is designed to capture, store, manipulate, analyse, manage and present spatial or geographic data. In other words, the data is in some way related to geographical locations. The use of artificial intelligence in conjunction with GIS has had a great impact on traffic management, ride sharing, etc. Let's look at some other applications.



Over the years, artificial intelligence (AI) has become a buzzword that symbolises the next stage of innovative technological transformations and how industry will be driven in the future. Using intelligent algorithms, data classification and smart predictive analysis, AI can be used in a number of sectors ranging from healthcare, finance, advertising and retail, to manufacturing and transport. During the last ten years, machine learning and AI have been rapidly transforming many areas related to GIS and spatial applications. Artificial intelligence provides sophisticated techniques for GIS projects, while GIS is a powerful technology with vast data sets and wide scope for the use of AI. For example, fuzzy logic has been successfully applied to imprecise spatial issues like data collection,

representation and analysis as well as for the classification of land, soil and remotely sensed imagery.

Machine learning and artificial intelligence in GIS

Technologies like AI, deep learning and machine learning (ML) have matured just in time to equip us with the tools we need to make sense of an ever-increasing trove of imagery data in the context of GIS. Simply put, ML makes sense out of noisy data, finding patterns that you'd never think existed. It empowers the geospatial ecosystem by providing real-time near-human level perception, integrates into analytical workflows, and drives data exploration and visualisation. It thus automates the entire process of creating scalable insights from large amounts of data. In other words, it is

software that writes software. Instead of applying a pre-built function, ML gains experience through repeatedly seen/experienced conditions and builds a model that can be applied in a new situation. Such machines are able to understand geospatial information, and with deep learning, they are able to obtain geospatial information from their surroundings, as required, and process it in real-time to do their jobs. For example, Google might use Bayesian classification to filter spam emails. Alternatively, Facebook might use it for facial recognition and automatically identify faces in images.

Types of machine learning

The two broad categories of machine learning that can apply to GIS applications in various ways are supervised and unsupervised learning.